Model Validation Report

Model Name: Mortgage Insurance Credit Risk Model
Model Version: [Insert Version]
Business Area: Mortgage Insurance Credit Risk
Validation Date: [Insert Date]
Validator: [Insert Name]
Model Owner: [Insert Name]
Review Period: [Insert Period]
Intended Use: Mortgage insurance credit risk assessment and underwriting support

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1. Purpose and Scope

Objective

To assess whether the development data are complete, accurate, relevant, representative, and appropriately prepared for the intended model use.

Validation Assessment

Review Item	Status
The model purpose has been clearly documented	Yes
The target population has been clearly defined	Yes
The development period is appropriate for the intended use	Yes
The data sources have been identified and documented	Yes
The scope of the review is appropriate for the model complexity	Yes

Comments

The review covered data loading, preprocessing, borrower aggregation, feature engineering, and target construction logic used for mortgage insurance credit risk modeling. The framework includes policy data, borrower information, credit bureau data, economic variables, and LRU Plus data.

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2. Data Source and Lineage Review

Objective

To confirm that the data lineage from source systems to the final analytical dataset is traceable and auditable.

Validation Assessment

Review Item	Status
All source systems have been identified	Yes
Data extraction dates have been documented	Partial
Source-to-target mapping has been provided	Partial
Record count reconciliation has been completed	Partial
Filters, joins, and exclusions are documented	Yes
Manual interventions are documented and justified	Partial
Final dataset can be reproduced	Yes

Comments

The framework loads data from multiple parquet and CSV files using centralized utility functions. The process includes borrower aggregation, economic data integration, SID correction logic, and duplicate removal. Datasets are merged using application and borrower identifiers.

Some reconciliation and extraction details should be documented more clearly for audit purposes.

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3. Data Completeness

Objective

To assess whether key variables are sufficiently complete for model development and validation.

Validation Assessment

Review Item	Status
Missingness has been quantified for key variables	Partial
Missingness reviewed by segment and geography	No
Missingness is within acceptable tolerance	Yes
Missingness patterns align with business expectations	Yes
No major concentration of missing values identified	Yes
Elevated missingness assessed for impact	Partial

Comments

The preprocessing logic includes multiple fallback methods to reduce missing values, especially for bureau scores and property values. Credit scores are derived using several alternative bureau sources when primary values are missing.

However, formal missing-value reporting by segment, geography, and vintage was not identified.

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4. Data Accuracy and Consistency

Objective

To confirm that data values are plausible, internally consistent, and aligned across systems.

Validation Assessment

Review Item	Status
Key variables tested for logical consistency	Yes
Date sequencing is valid	Yes
Derived fields reconcile to source fields	Partial
Duplicate records identified and addressed	Yes
Impossible values handled appropriately	Yes
Field definitions are consistent across systems	Partial

Comments

The framework applies SID correction logic, numeric validation, duplicate removal, and interest-rate normalization. Credit scores outside valid ranges are replaced with missing values. Loan amortization and contract-rate values are also validated and corrected where necessary.

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5. Data Relevance and Suitability

Objective

To determine whether the data are representative of the intended mortgage insurance portfolio and use case.

Validation Assessment

Review Item	Status
Development sample reflects intended population	Yes
Product types are appropriately represented	Yes
Channel mix is broadly consistent	Partial
Geographic coverage is appropriate	Yes
Vintage coverage is sufficient	Partial
Stress and benign periods represented	Partial
No major survivorship bias identified	Partial

Comments

The dataset includes mortgage insurance business, LRU Plus data, multiple regions, and economic variables such as HPI and unemployment rates. Regional mapping includes Toronto segmentation and rural classification.

Additional testing should confirm representativeness during stressed economic periods.

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6. Outcome Construction

Objective

To confirm that the dependent variable is correctly defined and constructed without leakage.

Validation Assessment

Review Item	Status
Outcome definition documented	Yes
Observation and performance windows specified	Partial
Event timing is appropriate	Yes
Censoring rules documented	Partial
No future information used in labels	Yes
Outcome assignment independently verified	Partial

Comments

Target labels are generated using funded and declined policy statuses. Additional logic is applied for LRU-funded classifications.

No obvious leakage issues were identified in the target-construction logic. However, observation and performance windows should be documented more clearly.

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7. Preprocessing Review

Objective

To assess whether preprocessing steps are documented, reproducible, and appropriate.

Validation Assessment

Review Item	Status
Preprocessing steps documented	Yes
Filtering rules justified	Yes
Deduplication logic appropriate	Yes
Transformations suitable for variable distributions	Yes
Binning and grouping reasonable	Yes
Scaling and normalization appropriate	Yes
No major preprocessing bias identified	Yes
Preprocessing pipeline reproducible	Yes

Comments

The preprocessing framework includes tenure mapping, property-age calculation, property-value calculation, economic-region assignment, contract-rate normalization, and range clipping.

The preprocessing structure is organized and configuration-driven.

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8. Missing Value Treatment

Objective

To assess the appropriateness of the missing data methodology.

Validation Assessment

Review Item	Status
Missing values treated consistently	Yes
Imputation rationale documented	Partial
Missingness indicators considered	Partial
Dropped observations justified	Yes
No future information used for imputation	Yes
Impact of missing treatment assessed	Partial

Comments

The framework uses fallback logic, grouped averages, and formula-based reconstruction to handle missing values. Property values and bureau scores are estimated using alternative data sources when necessary.

Centralized documentation for all imputation rules is recommended.

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9. Outlier Treatment

Objective

To determine whether outlier treatment is methodologically sound and business-justified.

Validation Assessment

Review Item	Status
Outlier thresholds documented	Partial
Consistent methodology used	Yes
Extreme values reviewed for validity	Yes
Capping methods justified	Partial
Observation removal justified	Yes
Impact on performance reviewed	Partial
Valid tail-risk observations preserved	Partial

Comments

The framework applies range clipping, numeric validation, and hard caps for variables such as credit scores, property values, and contract rates.

Documentation supporting selected thresholds should be improved.

10. Business Rules and Aggregation

Objective

To confirm that business rules used in aggregation and feature construction are appropriate.

Validation Assessment

Review Item	Status
Aggregation rules documented	Yes
Record consolidation logic correct	Yes
Exposure duplication ruled out	Yes
Date alignment logic appropriate	Yes
Loan-level mapping accurate	Yes
Manual overrides documented	Partial
Aggregation preserves economic meaning	Yes

Comments

Borrower-level data are aggregated to the application level using weighted averages, minimum and maximum values, and combined bureau calculations. The framework also calculates borrower counts and applies fallback bureau logic.

11. Bias and Representativeness

Objective

To assess whether the dataset exhibits bias that may affect model validity.

Validation Assessment

Review Item	Status
Sampling bias assessed	Partial
Selection bias assessed	Partial
Survivorship bias assessed	Partial
Reject inference considered	No

Review Item	Status
Exclusions documented and justified	Yes
Segment-level differences reviewed	Partial
Identified bias assessed for materiality	Partial

Comments

No major bias concerns were identified during the preprocessing review. However, additional analysis should be performed to assess reject inference, stress-period representation, and segment-level stability.

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12. Reproducibility

Objective

To confirm that the data preparation process can be independently reproduced.

Validation Assessment

Review Item	Status
Code or logic version controlled	Yes
Parameters documented	Yes
Input and output files identified	Yes
Run logs available	Partial
Final dataset can be recreated	Yes
Results reconcile to approved version	Partial

Comments

The framework uses centralized configuration management with YAML configuration files and reusable preprocessing modules. This supports reproducibility and consistent execution across runs.

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13. Validation Conclusion

Validation Assessment

Conclusion Item	Status
No material issues identified	No
Issues are not material to fit-for-use	Yes
Issues require remediation before reliance	Partial
Data are fit for use	Yes
Data are fit for use subject to conditions	Yes
Data are not fit for use	No

Conclusion

The preprocessing and data preparation framework is generally suitable for mortgage insurance credit risk model development and validation.

Strengths include:

- Structured preprocessing workflow
- Strong borrower aggregation logic
- Economic feature integration
- Multiple data-quality controls
- Reproducible configuration framework

Areas requiring improvement include:

- Missing-data reporting
- Leakage testing
- Documentation of observation windows
- Outlier threshold justification
- Stress-period representativeness analysis

Overall, the data are considered fit for use subject to the remediation and documentation improvements identified during validation.

Key Limitations

- Limited formal missing-data reporting
- Limited stress-period representativeness analysis
- Observation and performance windows require additional documentation
- Formal leakage testing was not identified

Required Remediation Actions

- Implement formal missing-data reporting
- Add leakage-testing procedures
- Improve preprocessing documentation

- Document outlier thresholds and rationale
- Validate stress-period representation

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14. Sign-off

Validator Name: [Insert Name]
Validator Title: [Insert Title]
Date: [Insert Date]

Model Owner Response: [Insert Response]

Approval Committee / Reviewer: [Insert Name]
Date: [Insert Date]

Sign-off Comments